“SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

RP-25-26J-375

Project Proposal Report (Draft)

Bangamuwage M.D

B.Sc. (Hons) Degree in Information Technology

Specializing In Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date 28/08/2025

Signature of the Co - supervisor: Date

ABSTRACT

The goal of this project is to develop an intelligent system that can track soil nutrients and make real-time fertilizer recommendations. It is especially made for Sri Lankan cinnamon cultivation. Conventional fertilizer application methods involve regular manual soil testing and set schedules. These frequently result in reduced crop yields, environmental damage, and nutrient imbalances. A precision agriculture solution is proposed by this study. It integrates machine learning algorithms (Random Forest and CatBoost) with Internet of Things sensors (JXBS-3001 for NPK, FC-28 for moisture, and DHT11 for temperature and humidity). This system will provide data-driven fertilizer recommendations in real time.

Multi-parameter sensors are connected to ESP32 microcontrollers in the system. By using the MQTT protocol, these sensors can transmit their data to a cloud-based dashboard. By utilizing soil-specific datasets and machine learning techniques, it is possible to train optimal nutrient levels on live sensor readings. They arrive at precise deficit percentages for every nutrient (N, P, K) and provide prompt guidance on fertilizer usage.

Keywords: IoT Sensors, Machine Learning, Precision Agriculture, Cinnamon Cultivation, Smart Farming, Nutrient Management

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LIST OF ABREVATIONS

Table I : List of Abbreviations

GPS	Global Positioning System
GIS	Geographic Information System
MERN	MongoDB, Express, React, Node.js
DSS	Decision Support System
SDSS	Spatial Decision Support System
SDLC	Software Development Life Cycle
API	Application Programming Interface
IDE	Integrated Development Environment
LKR	Sri Lankan Rupee
WBS	Work Breakdown Structure
ML	Machine Learning
IOT	Internet of Things
EC	Electrical Conductivity
PH	Potential of Hydrogen (soil acidity/alkalinity)
NPK	Nitrogen, Phosphorus, Potassium (primary plant nutrients)
DOA	Department of Agriculture (Sri Lanka)
ERP	Eppawala Rock Phosphate
SOP	Sulphate of Potash
MOP	Muriate of Potash

1. INTRODUCTION

Ceylon cinnamon (*Cinnamomum verum*) is one of Sri Lanka's most valuable spice crops, ranking as the country's second-largest agricultural export after tea. Despite cinnamon's high economic importance, traditional cultivation practices have led to suboptimal productivity and sustainability. Fertilizer applications are currently based on fixed schedules and occasional soil tests which often results in over-fertilization (causing nutrient leaching, soil acidification, and groundwater pollution) or under-fertilization (leading to poor growth and yields). Both scenarios harm farmers financially and environmentally. Excessive use of nitrogen fertilizers, for example, has been linked to significant soil acidification in intensive cropping systems. Surveys of cinnamon growers in Sri Lanka underscore these issues: the majority report rising fertilizer costs and uncertainty about proper application rates, with fertilizer expenses often not yielding proportional improvements in output [1]. In one recent survey, farmers noted that chemical fertilizer prices had become prohibitively high and often unavailable when needed. Furthermore, most smallholders lack access to regular soil testing - laboratory results take weeks to return, by which time field conditions may have already changed. Portable soil NPK test kits exist, but their adoption is low due to cost and limited awareness among farmers.

Overall, current fertilizer management for cinnamon is largely a one-size-fits-all approach that ignores field-specific conditions and real-time crop needs. [2] This context highlights an urgent need for a smarter fertilizer recommendation system that can provide precise, data-driven guidance tailored specifically to cinnamon cultivation.

1.1 BACKGROUND AND LITERATURE SURVEY

IoT in Precision Agriculture

Advances in precision agriculture are transforming how farmers manage crops and soil by leveraging the Internet of Things (IoT). IoT refers to networks of smart sensors and devices that collect real-time data from the field, which can be transmitted and analyzed to inform agricultural decisions. In crop production, IoT-based sensor networks monitor critical parameters such as soil moisture, temperature, pH, electrical conductivity, and nutrient levels continuously. These real-time measurements enable farmers to move from schedule-based to condition-based management of irrigation and fertilization. For instance, soil moisture sensors can trigger irrigation only when needed, preventing water waste, while pH and EC sensors help detect soil health issues early. By integrating field sensors with cloud platforms, farmers can remotely monitor conditions and receive alerts or recommendations on their mobile devices. IoT systems coupled with data analytics and artificial intelligence allow for predictive insights – for example, forecasting pest or disease outbreaks and predicting yield – so that interventions can be made proactively. Case studies from around the world have demonstrated that adopting IoT in agriculture leads to improved resource efficiency, higher yields, and reduced input waste. Mansoor *et al.* (2025) [3] review various smart farming deployments and note significant benefits such as 30% water savings from IoT-guided irrigation and more timely fertilizer applications, alongside challenges like high initial costs and connectivity issues. Indeed, challenges remain in IoT adoption for smallholders – including the upfront investment, the need for reliable rural internet connectivity, and the complexity of managing big data streams. However, the trend is clear that IoT provides a foundation for precision agriculture, shifting farm management from labor-intensive guesswork to data-driven decision-making. Even low-cost sensor setups have proven valuable: Idrissi *et al.* (2024) [4]developed an inexpensive wireless sensor network for a single plot, which successfully delivered real-time soil and weather data to farmers and improved their decision-making efficiency. Similarly, national programs and research initiatives are exploring IoT solutions for smallholders; for example, a review by Antony *et al.* (2020) [5] documented successful IoT implementations in low-income farming communities, which led to yield and quality improvements in crops like tomatoes through better microclimate control in greenhouses. These studies underscore the transformative role of IoT in agriculture: by providing continuous, site-specific information and automation capability, IoT enables farmers to apply inputs like water and fertilizers at the right place and right time – aligning with the "4R" principles (right source, right rate, right time, right place) of nutrient stewardship. In summary, IoT technology is a key enabler of precision farming, laying the groundwork for real-time monitoring and control that can be further enhanced with intelligent analytics.

Machine Learning for Fertilizer Recommendation

Building on IoT data streams, machine learning (ML) algorithms can analyze sensor readings and other farm data to generate intelligent fertilizer recommendations tailored to specific conditions. Traditional fertilizer guidelines often prescribe blanket rates for large regions, which may not reflect local soil variability. ML techniques offer a way to process multi-dimensional data (soil properties, weather, crop stage, etc.) and learn complex patterns that correlate with optimal fertilization outcomes. In recent years, researchers have developed various ML-based fertilizer recommendation systems with promising results. For example, Kollu *et al.* (2023) [6] implemented an IoT-driven framework that collected data on soil moisture, temperature, humidity, and weather, and used a Sequential Forward Selection algorithm with a multilinear regression model to recommend fertilizer dosage and timing. Their system achieved over 99% accuracy in recommending the correct fertilizer amount and time, outperforming traditional methods. Other studies have explored different ML models – from decision trees and random forests to neural networks – for nutrient management. Ennaji *et al.* (2023) [7] provides a comprehensive review of machine learning in nutrient management and reports that these models can successfully estimate soil nutrient levels and crop needs. A key advantage of ML-based systems is their ability to continuously improve as more data is gathered. The system by Kollu *et al.* could refine its regression model over time with additional sensor inputs, thereby increasing recommendation accuracy in future seasons. Transparency and interpretability of recommendations are also being considered. Musanase *et al.* (2023) developed an integrated crop and fertilizer recommendation system for Rwandan farmers which combined a neural-network crop selection model (achieving ~97% accuracy) with a rule-based fertilizer recommendation module. The rule-based component was built on expert agronomical knowledge, ensuring that the ML-driven suggestions aligned with sensible nutrient management rules for each crop. This approach yields recommendations that farmers and extension officers can trust, while still leveraging data-driven insights. Several prototype decision-support tools have targeted smallholder farmers. For instance, Sharma *et al.* (2025) introduced FertiCal-P, an Android-based app that takes soil test values (pH and N-P-K levels) as input and outputs site-specific fertilizer recipes for a given crop. The app uses a combination of rule-based logic and regression and even provides cost estimates for the recommended fertilizers, helping farmers make both agronomic and economic decisions. Initial trials in India showed that such digital tools can lower fertilizer usage and input costs while maintaining yields, by avoiding over-application. Likewise, Khan *et al.* (2022) [8]demonstrated an IoT-assisted, context-aware fertilizer recommendation system that first maps soil fertility in real time and then applies ML classifiers to suggest optimal nutrient applications. Among several ML models tested (Logistic Regression, SVM, k-NN, etc.), their system found Gaussian Naïve Bayes to be most effective, achieving about 94–96% accuracy in

recommending the proper fertilizer actions in field trials. These data-driven approaches consistently outperform static recommendations by accounting for local and temporal variability. Importantly, they also support the 4R nutrient stewardship by helping farmers apply the right nutrient source at the right rate, time, and place. Early results from on-farm experiments indicate that ML-informed fertilizer schedules can increase yields and nutrient use efficiency. In Ethiopia, Liben *et al.* (2024) showed that a site-specific ML-based fertilizer recommendation for wheat improved grain yields by 16–25% and profit by over $400 per hectare compared to the government's blanket recommendation, while also raising nitrogen use efficiency by 30%. However, they also caution that ML models are not infallible – their study noted a few sites where the recommendations led to lower phosphorus or sulfur efficiency, highlighting the need for continuous model refinement. Similarly, Tanaka *et al.* (2024) evaluated ML methods for predicting economically optimal fertilizer rates in Japan. They found that while ML (random forests, neural networks, etc.) could accurately predict crop yields, the models sometimes suggested fertilizer rates that deviated from traditional expert recommendations, especially when trained on limited data. This underscores the importance of agronomic domain knowledge and validation when deploying ML systems in the field. Despite these caveats, the overall trajectory in research is very encouraging: smarter fertilizer recommendation systems – powered by IoT data and ML algorithms – are emerging as a key innovation for sustainable agriculture. They enable farmers to move away from one-size-fits-all fertilization to more site-specific, real-time nutrient management. Notably, most of the current solutions focus on major staples like rice, maize, and wheat. There is a clear research gap when it comes to developing such systems specifically for high-value spices like cinnamon. No published studies to date have presented an IoT+ML fertilizer recommendation platform tailored to cinnamon's unique growth patterns and nutrient uptake. Addressing this gap will involve training models on cinnamon-specific data and calibrating sensor thresholds to the needs of cinnamon cultivation. By doing so, Sri Lanka's cinnamon farmers could achieve the "precision nutrition" of their crop: applying the right type and amount of fertilizer exactly when and where the cinnamon plants need it. This would reduce wasteful fertilizer use (preventing issues like soil acidification and nutrient runoff) and enhance yield and bark quality – ultimately improving both environmental sustainability and farmer livelihoods.

2. RESEARCH GAPS

To identify the research gap, four existing agricultural automation systems (denoted here as Research A, B, C, D) were compared against the key features required for a comprehensive smart farming solution. These features include operational flexibility, achievement of user-specific goals, accommodation of individual farm requirements, availability of customizable solutions, integration of AI/IoT technologies, and utilization of real-time data. **Table 3.1** provides an overview of this comparative analysis, indicating with a checkmark (✓) which features each system supports and with a cross (✗) where it falls short. As evident from the table, each prior study excels in certain aspects, but none covers all the features simultaneously, thereby leaving notable gaps in functionality and scope.

Table 3.1. Comparison of past studies (Research A–D) and the Proposed System across key features (✓ = feature present; ✗ = feature not supported).

Table I : research Gap

Feature	Research A	Research B	Research C	Research D	Proposed System
Flexibility	✓	✓	✗	✗	✓
Achieves Personal Goals	✓	✓	✓	✗	✓
Meets Individual Requirements	✓	✗	✗	✗	✓
Customizable Solutions	✗	✗	✗	✗	✓
AI/IoT Integration	✗	✗	✓	✗	✓
Real-Time Data Usage	✗	✗	✓	✓	✓

Research A focuses on automating land-clearance tasks and demonstrates high operational flexibility, successfully meeting broad user-defined goals in that domain. However, it provides limited support for specialized customization, lacking configurable solutions for unique farm needs. [9] Similarly, Research B employs GPS-based automation for fence installation to achieve efficient and accurate boundary setup. This system meets its niche objective, but it fails to fully accommodate individual farm requirements and does not

utilize real-time data feedback for adaptive adjustments. [10] In both cases, the automation is largely task-specific; while effective within their own scope, these systems lack the adaptability to be repurposed or tailored beyond their primary functions.

Research C offers a more comprehensive farm management approach by integrating AI analytics and real-time sensor data, which enhances decision-making precision and responsiveness. [11] This advanced system addresses some challenges by leveraging data-driven, AI-supported interventions (for example, optimizing irrigation or crop treatments in real time). Yet, it still lacks a broad range of configurable modules to adapt to different agricultural scenarios, limiting its flexibility in meeting diverse operational requirements. Research D, on the other hand, concentrates on incorporating weather forecasting into farm operations, providing farmers with crucial real-time climate information (e.g., localized temperature and rainfall predictions) [12] for proactive planning and risk mitigation. Nevertheless, because it does not integrate advanced AI or broader IoT-based automation, Research D cannot automatically respond to the changing conditions it monitors. In effect, it misses the benefits of combining live data streams with intelligent control; such integration of real-time data and adaptive AI models is considered essential for handling unpredictable farm environments.

In summary, each of these existing studies (A -D) exhibits specific limitations whether in customizability, scope of automation, or intelligent data utilization that prevent them from fully satisfying the spectrum of smart farming needs. The Proposed System is designed to bridge these gaps by combining the strengths of prior works while overcoming their individual limitations. As evidenced in Table 3.1, the Proposed System concurrently achieves a high degree of flexibility, aligns with personalized farmer goals, and supports unique farm requirements through modular, customizable solutions. It integrates AI and IoT components throughout its architecture, continuously leveraging real-time data streams for data-driven decision-making and automated control across various agricultural tasks. This unified, intelligent platform effectively merges functions that were previously isolated in studies A - D - from autonomous field preparation and infrastructure deployment to dynamic farm management and environmental monitoring - into a single cohesive system .Such a holistic approach ensures that the platform can adapt to diverse farming scenarios and on-the-fly changes, providing data-driven solutions tailored to individual farm needs while also promoting sustainability . In doing so, the Proposed System addresses the unmet needs identified in the existing literature and represents a significant step toward a more flexible and smart farming paradigm.

3. RESEARCH PROBLEM

Cinnamon farmers in Sri Lanka face significant challenges with current fertilizer application practices. At present, fertilizers are applied on fixed schedules recommended by the Department of Agriculture. These guidelines provide general direction but do not account for real-time soil nutrient conditions, leading to inefficiencies: in some cases over-fertilization causes nutrient leaching and soil degradation, while in other cases under-fertilization reduces plant growth and bark quality. Both outcomes waste resources and harm long-term soil health. In short, applying fertilizer in quantities greater or less than the crop requires is economically and environmentally detrimental. Farmers bear unnecessary costs when over-fertilizing and sacrifice yield when under-fertilizing, showing that the status quo is suboptimal.

A field survey among cinnamon growers highlighted the extent of the problem. Over 75% of farmers reported dissatisfaction with current fertilizer practices, citing high input costs and uncertainty about whether the amounts they apply match their soil's actual nutrient needs. Most farmers use chemical fertilizers such as urea, [13] TSP, and MOP, yet these are often applied without any soil testing beforehand. This inefficient "blind application" means farmers are essentially guessing – some apply excess fertilizer "just in case," while others reduce application to save costs, neither of which optimizes the yield or quality of cinnamon.

Another major challenge is the lack of monitoring and timely feedback. The majority of smallholder farmers do not have regular access to soil testing laboratories. When they do send soil samples for analysis, the results often take weeks to return, by which time the opportunity for corrective fertilization may have passed. Sri Lanka's public agricultural labs are few and lack the capacity to serve all farmers quickly, resulting in 3–4-week delays for soil test results. As a result, most farmers skip testing altogether – only a small fraction submit samples due to the burden of transporting soil to distant labs and waiting extended periods for feedback. Portable NPK soil test kits exist as an alternative for field use, but adoption remains low. Many farmers are unaware of them or doubt their reliability, as these kits often provide qualitative, color-based readings that are difficult to interpret without technical training. Consequently, farmers lack any practical on-site tool to check their soil's nutrient status in real time.

Rising input costs further compound the issue. Fertilizer is one of the largest expenses in cinnamon cultivation, and prices have been highly volatile in recent seasons. Surveys indicate that over 80% of farmers reported difficulties in managing fertilizer costs, especially when yield increases were not proportional to the investment. Recent policy shifts, such as the temporary ban on chemical fertilizers in 2021 [14] and subsequent market disruptions, have caused fertilizer prices to skyrocket, placing them "beyond the reach of ordinary cinnamon farmers" in some cases. Many farmers now struggle to afford the recommended fertilizer amounts, and some cut down on applications, further reducing yields. The consensus among growers is that blanket fertilizer recommendations are not adapted to their specific soil conditions, leading to wasteful spending and reduced

profitability. In short, farmers feel they are spending more on fertilizer than they should, without realizing equivalent benefits in cinnamon growth or harvest.

These challenges underline an urgent need for a comprehensive, site-specific fertilizer recommendation system tailored to cinnamon. An ideal solution would integrate IoT-based soil sensors (pH, electrical conductivity, moisture, temperature) with nutrient sensors (portable NPK or nitrate sensors) to continuously monitor soil fertility. The data could then feed into machine learning models trained on cinnamon-specific datasets, enabling real-time analysis and precise fertilizer recommendations for each plot. [15] Such a system would provide farmers with site-specific, up-to-the-minute guidance on nutrient management. Research in other crops has already demonstrated that combining IoT sensors with AI models enables real-time decision support for farmers. Extending this capability to cinnamon would empower growers to move from guesswork to data-driven nutrient management.

In summary, the multitude of issues faced by cinnamon farmers – from inefficient fertilization and lack of soil data to high costs – calls for a holistic precision agriculture solution. The proposed system should not only deliver accurate fertilizer recommendations but also include land management features (e.g., mapping cinnamon plots and tracking soil health), and economic analysis tools to balance costs and returns. This integrated approach would help cinnamon farmers optimize fertilizer use, increase productivity, and improve profitability. Ultimately, empowering farmers with precise measurements and AI-driven recommendations will ensure more sustainable and successful cinnamon cultivation in Sri Lanka.

4. OBJECTIVES

4.1 Main Objectives

To develop and deploy an intelligent, real-time soil nutrient monitoring and fertilizer recommendation system that optimizes cinnamon cultivation through precision agriculture technologies, machine learning analytics, and IoT sensor networks

4.2 Specific Objectives

4.2.1 Real-time nutrient gap analysis

One of the primary objectives of this research is to develop a system that provides accurate and real-time analysis of soil nutrient levels in cinnamon fields. Using IoT-based soil sensors (pH, EC, moisture, temperature) integrated with periodic NPK measurements, the system will calculate the percentage deficit for each key nutrient (Nitrogen, Phosphorus, Potassium). Unlike traditional methods, which rely on periodic lab tests, this approach enables farmers to instantly identify nutrient imbalances and respond accordingly, ensuring more efficient fertilizer use and preventing yield loss.

4.2.2 Dynamic fertilizer dose calculation

The system will incorporate algorithms that convert nutrient deficits directly into fertilizer dose recommendations. For example, if the soil is 20% deficient in nitrogen, the system will translate this into an exact fertilizer application amount (e.g., 25 kg Urea/acre). This ensures that farmers receive actionable and site-specific recommendations, tailored to each plot's soil condition. The tool aims to reduce overuse and underuse of fertilizers while improving cost efficiency for cinnamon cultivation.

4.2.3 Machine learning–driven recommendation engine

The project will integrate Machine Learning models such as Random Forest and CatBoost, trained on cinnamon-specific soil and yield datasets. These models will enhance the accuracy of fertilizer recommendations over time by learning from past soil data, farmer feedback, and yield outcomes. By doing so, the system evolves into an intelligent advisor that continuously improves its recommendations for different regions and field conditions.

4.2.4 Cloud-based monitoring and decision support

To support large-scale adoption, the system will include a cloud-based dashboard where plantation managers can view real-time soil conditions, nutrient deficits, and fertilizer prescriptions across multiple estates. The dashboard will also store historical records, generate reports, and provide alerts for critical nutrient shortages. This functionality enables data-driven decision-making at both field and enterprise levels,

making the tool valuable for both smallholder farmers and export companies managing multiple plantations.

5. METHODOLOGY

5.1 SYSTEM OVERVIEW DIAGRAM

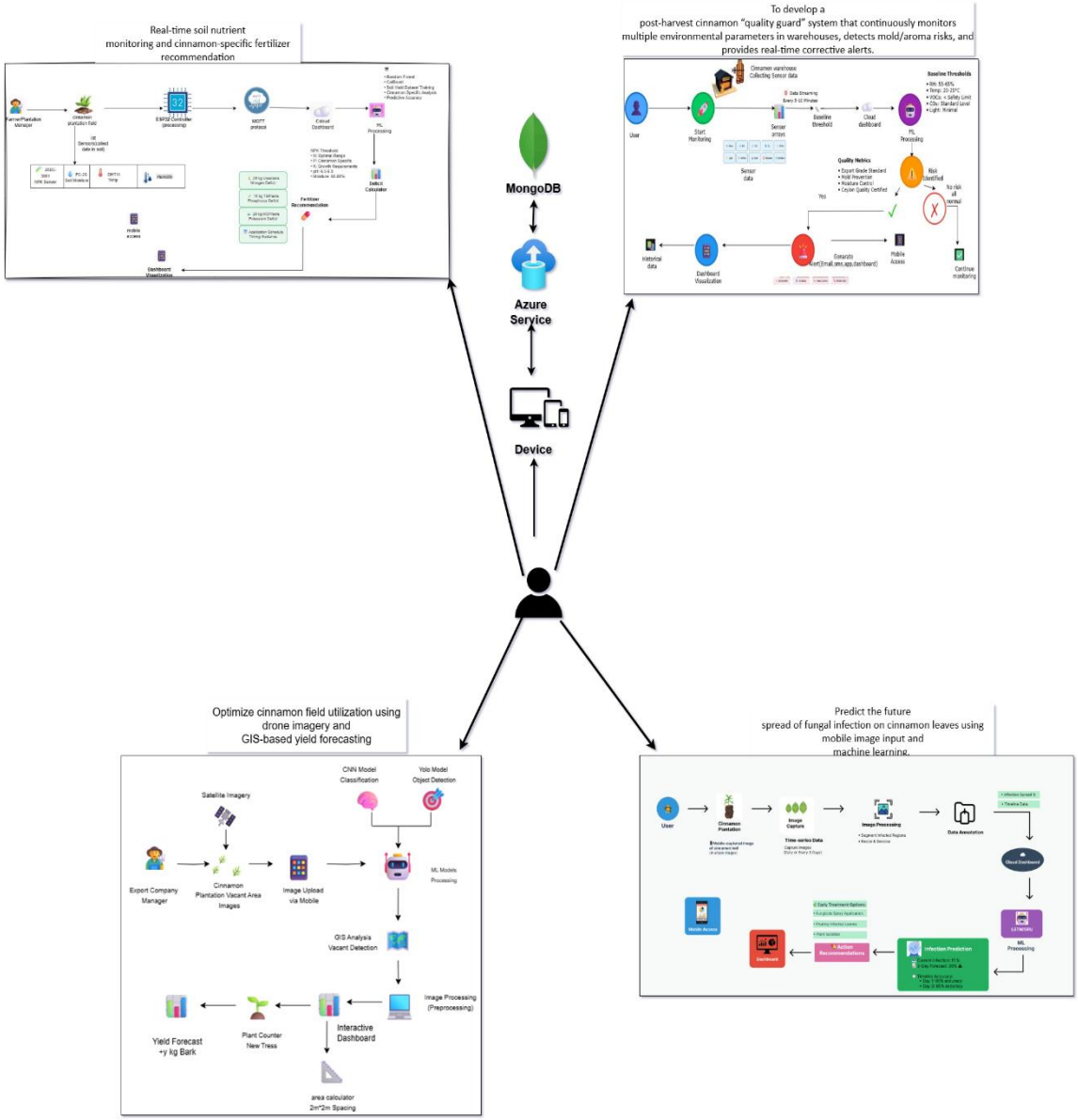

Figure 1 : Component System Diagram

SMARTCINNAMON is an integrated AI–IoT–GIS platform for the cinnamon value chain. Field and warehouse devices stream data to Azure services and MongoDB; machine-learning models turn that data into real-time recommendations, risk alerts, forecasts, and dashboards delivered on mobile and web.

At the center is the user's device (mobile/web) connected to cloud services. IoT sensors in fields and warehouses send readings to Azure (via MQTT/REST) and are stored in MongoDB. Cloud ML services analyze the incoming data continuously and return insights to dashboards and alerting channels. The platform combines four modules that work together to raise yield, protect post-harvest quality, and control disease.

Real-time Soil & Fertilizer Intelligence - ESP32 nodes read NPK (JXBS-3001), soil moisture (FC-28), and temperature/humidity (DHT11). Live values are compared with cinnamon-specific target ranges. ML models (e.g., Random Forest, CatBoost) learn from soil + yield history to recommend exact fertilizer doses (e.g., kg/acre), showing a percent "deficit" for N, P, and K and converting that into actionable inputs.

Post-harvest "Quality Guard" - Multi-sensor stations monitor 10 factors inside warehouses: Temperature, Relative Humidity, CO_2, O_2, VOCs, light intensity, airflow, dust/particles, vibration/shock, and surface moisture. Baselines (e.g., RH 55-65%, 20-25 °C) and ML analytics detect mold/aroma risk early, trigger corrective guidance (ventilate, dehumidify, reduce light), and keep audit-ready logs and trends.

GIS Field Planner & Yield Forecast - Drone/satellite imagery is segmented to map cultivated vs. vacant areas. The system proposes optimal replanting layouts using standard spacing (e.g., 2 m × 2 m), estimates additional plant counts, and combines tree counts with historical bark productivity to forecast yield. Results appear on an interactive spatial dashboard: "X new trees → +Y kg expected yield."

Leaf Disease Detection & Spread Prediction - A mobile photo of a cinnamon leaf is analyzed by a CNN to detect current infection and severity. Time-series models (LSTM/GRU), trained on annotated progressions, predict how the infection will grow over the next few days (e.g., "+18% in 72 h"). The app visualizes the forecast and recommends early actions such as pruning, targeted spraying, or isolating plants.

Data Flow - Sensors/Images → Azure ingest → MongoDB storage → ML analytics → Dashboards & Alerts → Farm/warehouse actions.

Value - The platform delivers precise fertilizer dosing, proactive post-harvest quality control, optimized land utilization with yield gains, and early disease intervention-

improving productivity, compliance, and sustainability across the cinnamon supply chain.

5.2 COMPONENT SYSTEM DIAGRAM

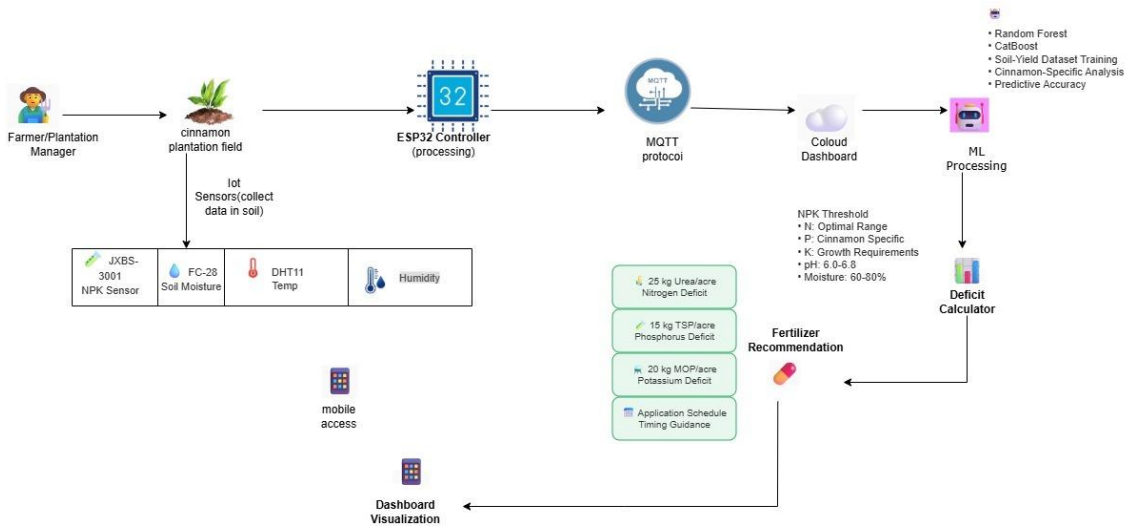

Figure 2 : Component System Diagram

According to the above diagram, users can monitor soil nutrients and receive precise fertilizer recommendations through an integrated IoT and machine learning–based system. The design is user-friendly and suitable for both smallholder farmers and large cinnamon export companies. It contains several features that improve the usability, accuracy, and efficiency of fertilizer management.

- **Data Collection:** IoT devices (pH, EC, moisture, soil temperature sensors) are deployed in selected field plots. Supplementary NPK test kits and nitrate sensors are used periodically to calibrate sensor readings.

- **Processing Layer:** The ESP32 microcontroller collects data from sensors and transmits it via Wi-Fi/LoRa to the cloud. Machine learning models (Random Forest, CatBoost) process the data, perform nutrient gap analysis, and calculate the percentage deficit for Nitrogen, Phosphorus, and Potassium.

- **Recommendation Engine:** The nutrient deficit is converted into actionable fertilizer prescriptions (e.g., “Apply 25 kg Urea and 10 kg ERP per acre”). These recommendations are aligned with Department of Agriculture guidelines but dynamically adapted to real-time soil conditions.

- **Dashboard & User Interaction:**

- Farmers and estate managers can view soil status, nutrient gaps, and fertilizer prescriptions via a web-based dashboard.
- Reports can be generated to track input costs, field performance, and yield improvements.
- A bilingual (English/Sinhala) interface ensures accessibility for field staff and company managers.

This component system ensures that cinnamon fields are managed with precise, data-driven fertilizer applications. By providing real-time nutrient monitoring and instant fertilizer recommendations, the system reduces waste, improves crop yield, and supports sustainable agricultural practices.

5.4 PROJECT TECHNOLOGY STACKS

Table II : Project Technology Stack

Tools & Technologies	Purpose
React-Native	For mobile application frontend implementations.
React.js	For web application frontend implementations.
Node.JS	For backend implementation
Python	Machine learning-related implementations.
Scikit-learn/CatBoost/ Random Forest	Training and running ML models for fertilizer recommendation
MongoDB/Firebase	Use As Database
Portable NPK Kit / Nitrate Sensor	For periodic nutrient calibration and validation of sensor data.
Google Maps API	For accessing GPS data and map functionalities. (Location API, MAP API)

Grafana / Recharts	Grafana / Recharts.
Visual Studio Code	Use As IDE
Git	Use Git as the version controller.
Postman	Use postman for test API requests.
Azure	As a project management tool

5.5 SOFTWARE SOLUTION

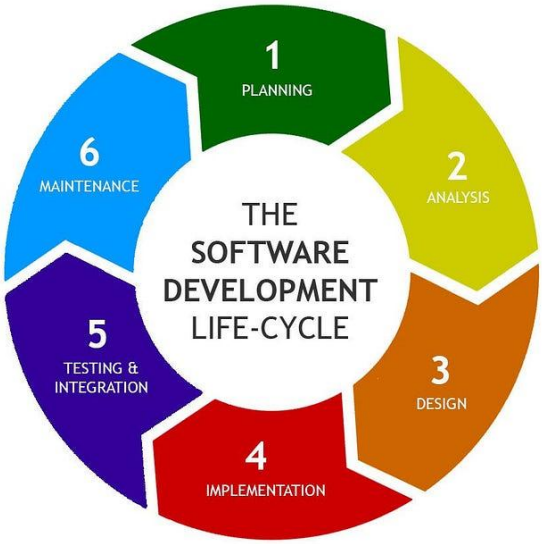

Figure 3 : SDLC Life Cycle

We will follow the Agile Software Development Life Cycle (SDLC) model to guide our project. Agile provides a flexible and iterative approach, allowing us to plan, develop, test, and deliver high-quality software in stages. This methodology encourages continuous improvement, frequent user feedback, and quick adjustments, ensuring the final product meets user needs effectively. By breaking the project into manageable sprints, the team can focus on delivering specific features incrementally, leading to a more efficient and responsive development process. This approach will help us build a robust and user-friendly land management application.

6. PROJECT REQURMENTS

6.1 FESIBILITY STUDIES

Conducting a feasibility study is essential to determine the ability to develop the proposed Smart Fertilizer Recommendation System for Cinnamon Fields. The study evaluates the project from four perspectives: technical, operational, economic, and schedule feasibility.

Technical feasibility: The project is technically possible since it makes use of well-established and widely documented technologies. IoT hardware such as ESP32 microcontrollers and sensors for soil pH, EC, moisture, and temperature are commercially available and affordable. Supplementary nutrient detection tools (e.g., portable NPK test kits and nitrate sensors) will be integrated to calibrate sensor data. On the software side, the system will be built using Node.js/Express for the backend, MongoDB for data storage, and React.js for the front end. Machine learning models (Random Forest, Decision Trees, Neural Networks) can be implemented in Python using Scikit-learn and TensorFlow. The development team possesses the necessary knowledge and technical expertise to successfully apply these technologies.

Operational feasibility: The system is designed to meet the needs of smallholder cinnamon farmers as well as larger estate owners, offering real-time insights into soil nutrient status and fertilizer requirements. The dashboard will provide simple, bilingual (English/Sinhala) recommendations to ensure accessibility to farmers with limited technical knowledge. Feedback loops will allow farmers to report results (yield, plant health), which will be used to refine the system. Early field testing with selected cinnamon plots in Matara, Galle, and Kalutara districts will validate usability and operational requirements.

Economic feasibility: The project is economically viable, with a clear path to revenue generation. The business model will combine subscription-based advisory services with hardware kits supplied to farmers. Initial costs per site (IoT sensor kit + NPK kit) are estimated at LKR 50,000–60,000, with an annual cloud hosting and service cost of LKR 25,000. For farmers, the expected cost per month is significantly lower than their current fertilizer losses due to inefficiency. Considering Sri Lanka's cinnamon industry exports over US$200 million annually, improving fertilizer efficiency and yield even by 10–15% could generate substantial economic benefits at the sector level. Thus, the return on investment for both farmers and developers is strong.

Schedule Feasibility: The project will follow the Agile Software Development Life Cycle (SDLC), enabling flexible and iterative development. The first three months will

focus on sensor integration, database setup, and dashboard development. The following months will cover dataset creation, ML model training, and field validation trials. A pilot system can be deployed within 6–8 months, with a targeted release aligned with the Maha planting season (late 2025) to ensure maximum field adoption.

6.2 USER REQUIREMENTS

- Enable accurate monitoring of soil parameters (pH, EC, moisture, temperature).
- Provide nutrient deficiency detection and site-specific fertilizer recommendations (type and quantity).
- Generate centralized dashboards and reports for plantation managers to track soil health and fertilizer usage.
- Integrate cost tracking and resource management tools to help exporting companies manage input expenses effectively.

6.3 FUNCTIONAL REQUIREMENTS

- Provide accurate soil data collection using IoT sensors (pH, EC, moisture, temperature).
- Offer real-time nutrient deficiency detection (N, P, K) and calculate percentage-based deficits.
- Converting nutrient deficits into dynamic fertilizer does recommendations (e.g., kg per perch/acre).
- Integrate machine learning models (Random Forest, CatBoost) to improve recommendations over time.
- Display fertilizer prescriptions, soil trends, and reports via a web-based dashboard (with optional mobile access).
- Allow users (farm managers/export companies) to download reports and track input costs versus yield outcomes.
- Enable feedback entry from field results (e.g., yield, plant health) to retrain and enhance model accuracy.

6.4 NON-FUNCTIONAL REQUIREMENTS

- **Reliability**: 99.5% system uptime with redundant data collection
- **Scalability**: Support for 1000+ concurrent sensor nodes
- **Performance**: <2 second response time for dashboard queries
- **Security**: Encrypted data transmission and secure user authentication
- **Usability**: Intuitive interface design for farmers with limited technical experience
- **Compatibility**: Cross-platform mobile app (Android/iOS)

- **Accuracy**: >85% fertilizer recommendation accuracy validated through field trials

6.5 HARDWARE REQUIREMENTS

- GPS Enable Android Phone or iPhone with stable internet connection
- Laptop or Desktop with stable internet connection

- ESP32 microcontroller with Wi-Fi/LoRa capability
- Soil sensors: pH sensor, EC sensor, soil moisture sensor, soil temperature sensor
- Portable NPK test kit / handheld nitrate sensor (for calibration and validation)
- Laptop or Desktop with stable internet connection (for managers and supervisors)

6.6 SOFTWARE REQUIREMENTS

- Mobile Operating System: The application should be compatible with the latest versions of iOS and Android.
- Desktop/Laptop: Windows 10/11, macOS

6.6.1 BACKEND INFRASTRUCTURE

- Database: MongoDB is required for storing user data, plantation, and fertilization records.
- Backend Framework: Node.js for handling server-side logic and APIs.
- Machine Learning Libraries: TensorFlow for training and deploying ML models.

6 6.2. DEVELOPMENT TOOLS

- **Arduino IDE**: ESP32 programming
- **Python 3.8+**: Machine Learning Model Development
- **React Native**: Mobile application development
- **Node.js**: Backend API development
- **PostgreSQL/Influx DB**: Database systems

6.8 TEST CASES

Table III : Test Cases

Test Scenario	**Soil Data Collection with IoT Sensors**
Test Scenario ID	**T1**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify sensor readings are captured correctly from pH, EC, moisture, and temperature sensors	• ESP32 and sensors deployed in field. • Stable internet connection.	Soil condition values detected by sensors.	System displays accurate real-time sensor readings on dashboard.	Data is stored in database for analysis and reporting.

Test Scenario	**Nutrient Deficiency Detection**
Test Scenario ID	**T2**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify the system identifies nutrient deficits (N, P, K) from sensor + kit data.	nutrient deficits (N, P, K) from sensor + kit data. • Soil sensors and portable NPK kit available. • User uploads calibration data.	Sensor readings + NPK kit results.	System calculates % nutrient deficit (e.g., Nitrogen - 20%).	Nutrient deficit logged in database for recommendation engine.

Test Scenario	**Fertilizer Dose Calculation**
Test Scenario ID	**T3**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify nutrient deficit is converted into a fertilizer recommendation.	• ML model trained with soil–fertilizer dataset.	N deficit = 20%, P deficit = 10%.	Recommendation generated (e.g., Apply 25 kg Urea + 10 kg ERP per acre).	Prescription saved and displayed on dashboard.

Test Scenario	**Dashboard Visualization**
Test Scenario ID	**T4**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that soil data, trends, and recommendations are displayed correctly.	• User logged into web dashboard. • Valid sensor and ML outputs available.	Select field/plot view.	Dashboard shows charts for pH, EC, moisture, nutrient gaps, and fertilizer prescriptions.	User can download report in PDF/Excel.

Test Scenario	**Feedback Logging & Model Update**
Test Scenario ID	**T5**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify farmer feedback is logged for future ML retraining.	• User logged in. • Dashboard accessible.	Enter yield data and field observations	Feedback stored in database and linked to soil/fertilizer record.	Data available for ML retraining to improve future recommendations.

7. DESCRIPTION OF PERSONAL AND FACILITIES

Table IV : Description of personal and facilities

Registration No	Name	Task Description
IT22592606	Bangamuwage M.D	• Analyze soil nutrient management challenges in cinnamon plantations. • Collect soil sensor data (pH, EC, moisture, temperature) and field requirements from cinnamon exporters. • Implement IoT-based soil monitoring system using ESP32 and sensors. • Develop algorithms for nutrient deficit calculation and fertilizer dose recommendation. • Train and validate ML models (Random Forest, CatBoost) using soil–yield datasets. • Design and implement cloud-based dashboard for real-time soil monitoring and fertilizer recommendations. • Develop reporting features for cost tracking and multi-location plantation management.

8. GANT CHART

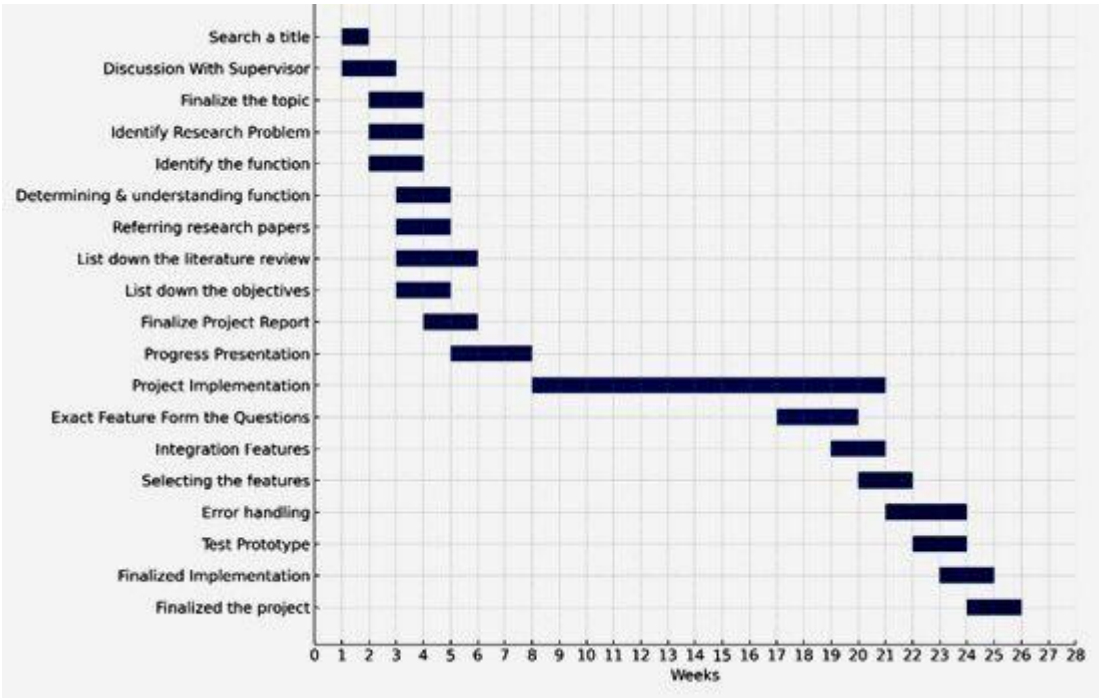

Figure 4 :Gant Chart

9. WORK BREAKDOWN CHART

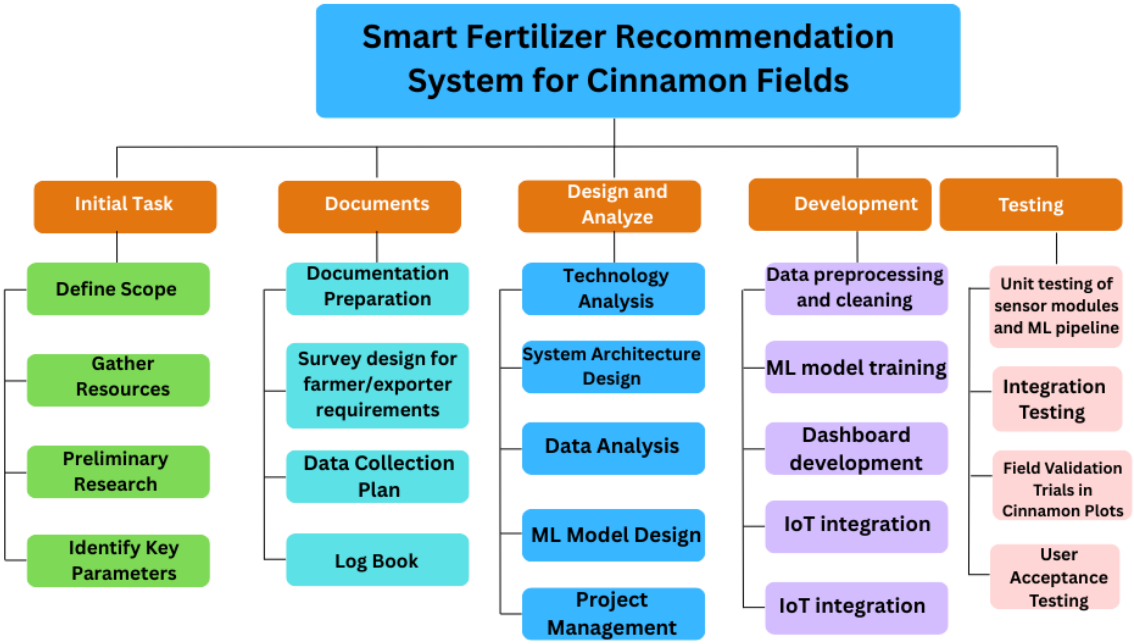

Figure 5 : WBS Diagram

10. COMMERCIALIZATION PLAN

10.1. BUSINESS PLAN

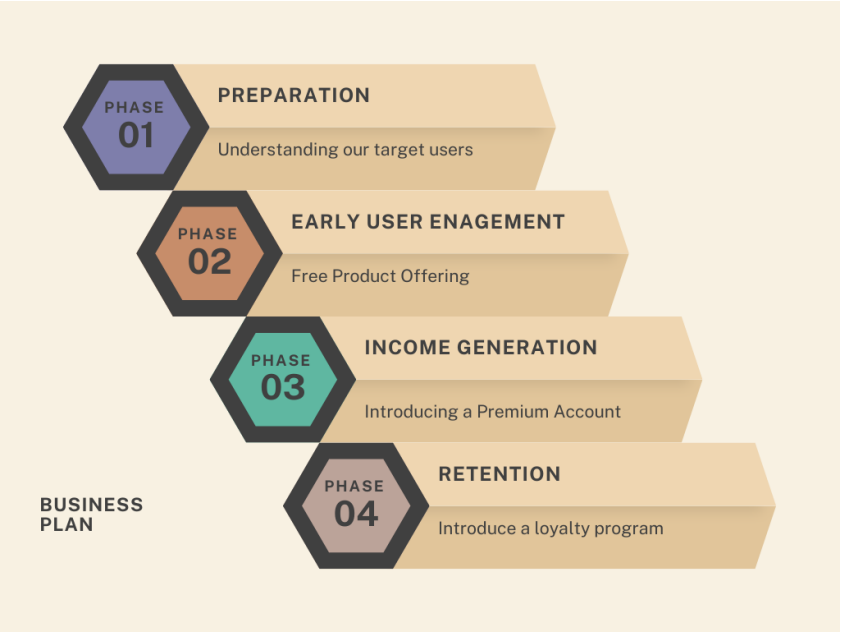

Figure 6 : Business Plan

Phase 01 - In this phase, we will prepare everything needed before launching the product. This includes understanding our target users and making sure the application works perfectly.

Phase 02 - We will offer the product for free to early users, especially in our target market. This is important to get feedback from farmers and land managers about how well the app works for them. Their feedback will help us make the app even better.

Phase 03 - After improving the app based on feedback, we will begin charging for some advanced features. These might include detailed financial tools and extra storage options. The basic paid plan will start at 400 LKR per month, which is affordable for most small and medium-sized farms. We hope to encourage users who tried the free version to upgrade to a paid plan for added benefits.

Phase 04 - In this final phase, we will focus on keeping our customers happy and encouraging them to stay. We will phase out the free version and introduce a loyalty

program where users earn points for sticking with the service or referring others. These points can be used for discounts or extra features, helping to build long-term customer loyalty.

10.2 MARKETING PLAN

SOCIAL MEDIA MARKETING

We will use social media to spread the word about our app. Our goal is to make people aware of the app, get them interested, and encourage them to subscribe. We'll create simple, helpful content like how-to videos, stories from real users, and tips on using the app. We'll focus on platforms that our target audience uses, like Facebook, Instagram, and WhatsApp.

SALES APPROACH

Our team will reach out directly to small and medium-sized farms. We'll do this through farming groups, local events, and online webinars where we can show how the app works and how it can help solve common farming problems.

CREATING CUSTOMER PACKAGES

We will offer different subscription plans, from basic to premium. Each plan will have different features, so users can choose what best fits their needs and budget. We'll encourage users to move from the free plan to a paid plan by offering more value with the higher-tier packages

11. BUDGET AND BUDGET JUSTIFICATION

Description	Amount (LKR)
IoT Sensors (NPK JXBS-3001, Moisture FC-28, Temp/Humidity DHT11, pH/EC Probes)	40,000
ESP32 Microcontrollers + Wi-Fi/LoRa Modules	15,000
Cloud Hosting & APIs (Azure, MQTT broker, Firebase/MongoDB Atlas)	25,000
ML/AI Model Training Infrastructure (Cloud GPU credits / Colab Pro / Azure ML)	20,000
Drone/Satellite Imagery (GIS Yield Planner)	10,000
Mobile Application Development (Fungal Spread Predictor)	Free
Multi-Sensor Array for Post-Harvest Quality Guard (Temp, RH, CO_2, O_2, VOC, Light, Airflow, Dust, Vibration, Surface Moisture)	50,000
Dev Tools & Software (VS Code, Postman, GitHub, GIS software licenses, testing tools)	Free
Miscellaneous (Travel to plantations, field trials, calibration, replacements)	10,000
Total Estimated Budget	170,000

This research addresses critical challenges in cinnamon cultivation through the development of an intelligent, IoT-based fertilizer recommendation system. By combining real-time soil monitoring with machine learning analytics, the project offers a comprehensive solution to optimize fertilizer application, reduce environmental impact, and improve crop yields.

The novelty lies in creating the first cinnamon-specific precision agriculture platform that provides real-time, data-driven fertilizer recommendations. Unlike generic solutions, this system is tailored to cinnamon's unique nutritional requirements and growing conditions in Sri Lanka.

Expected outcomes include significant improvements in fertilizer efficiency, crop productivity, and environmental sustainability. The commercial potential is substantial, with opportunities for technology transfer to other crops and international markets.

The project contributes to Sri Lanka's digital agriculture transformation while supporting the sustainability and competitiveness of one of the country's most valuable export crops. Through precision agriculture technologies, this research paves the way for more intelligent, sustainable farming practices that benefit farmers, consumers, and the environment.

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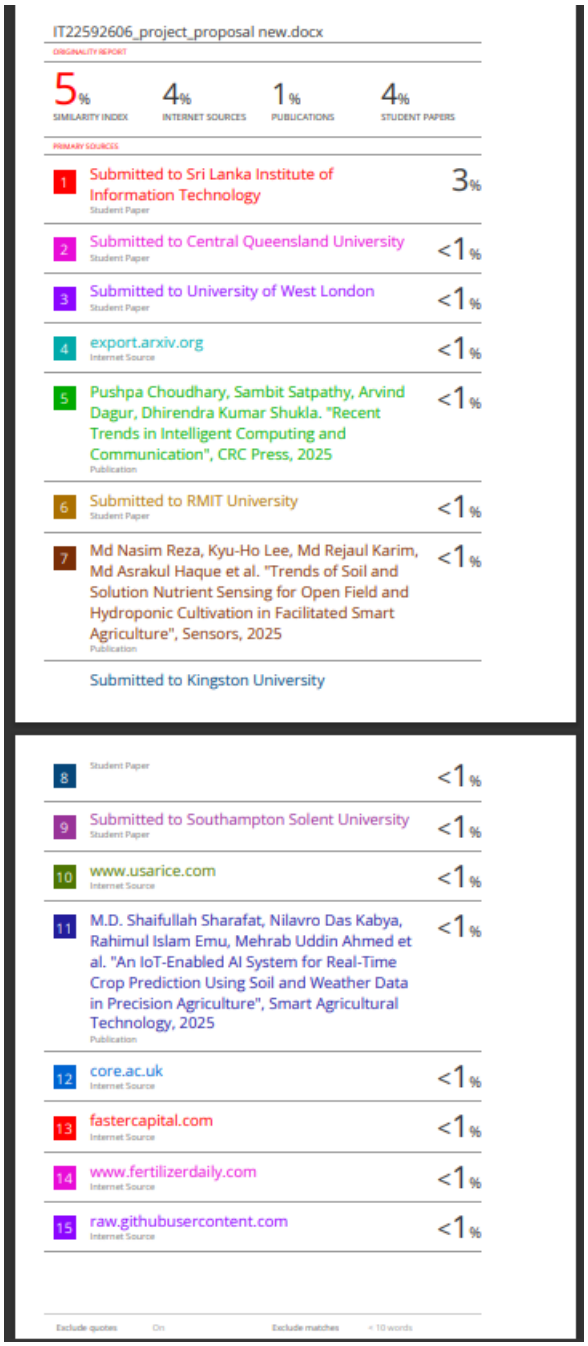
14. APPENDICIES



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